

SANDEEPKUMAR M D

Backend Engineer | Full Stack Engineer | Tiruppur, Tamil Nadu, India | Open to Relocation | Immediate Joiner

+91-7339680980 | mdsandeepkumar12@gmail.com | [linkedin.com/in/sandeepkumar-md](https://www.linkedin.com/in/sandeepkumar-md) | github.com/sandeep-md-12 | sandeepkumarmd.vercel.app

PROFESSIONAL SUMMARY

Backend and Full Stack Engineer with 1.8 years of experience taking product features from requirements to production with Python (FastAPI, Django), Node.js, React, PostgreSQL and AWS. Practices spec-driven development with AI tools (Claude, Cursor, Kiro), turning requirements into clear specs before building software that solves real problems for the teams who use it.

EXPERIENCE

Backend Developer (Freelance) | HLM - Human Resource Management Platform

Remote | May 2026 - Sep 2026

- Designed and shipped FastAPI REST APIs and microservices on Python and PostgreSQL with JWT authentication, Redis and Docker Compose, powering HR workflow modules for a platform serving 12,000 employees.
- Owned the new medical insurance module end to end: a 5-stage workflow that lets employees enrol, accept coverage and download e-cards and lets HR manage premiums and upload e-cards, with PostgreSQL stage tracking and AWS S3 file storage.
- Built the new reward disbursement feature: a category-based workflow where managers evaluated, scored and approved contributions, plus APIs letting employees redeem points for payouts or vouchers across 2 redemption options.
- Built a reusable push-notification service so every HR module could alert employees without extra backend work, queuing alerts in Redis and delivering via Firebase Cloud Messaging; turned requirements into specs before coding.

Software Engineer | Avasoft

Chennai, Tamil Nadu | March 2025 - January 2026

- Cut the operations team's Opstream invoice processing time by 40% for ~2,000 purchase orders a week with a new feature normalizing 5 ERP-specific JSON formats into one PostgreSQL schema, so they work from one format, not five.
- Shipped new Racpad features for 1,000+ daily staff users, including parcel pickup/drop-off assignment and feature-flagged delivery receipts in US English and Mexican Spanish for controlled release, on React, Node.js microservices and JWT.
- Wrote Jest unit tests with mocked dependencies for Racpad's Node.js APIs in Agile Scrum sprints.

Software Engineering Intern | Avasoft

Chennai, Tamil Nadu | September 2024 - February 2025

- Owned the new Absence Tracker end to end, giving 500+ employees one workflow for leave requests and approvers a multi-stage approval and status view.
- Built git-warden, a cross-platform Go CLI pre-commit hook that scans staged Git diffs with entropy analysis and regex rules for 50+ credential patterns (AWS, OpenAI, Stripe, Slack) and blocks commits that leak secrets.
- Implemented AWS CloudWatch logging and monitoring, improved GitLab CI/CD pipelines, built React and Node.js features and refactored 8 React components for maintainability.

PROJECTS

PG Management SaaS | FastAPI, PostgreSQL, React, Alembic, pytest

- Built a multi-tenant platform for hostel/PG owners, wardens and tenants that digitizes rent collection, with an income dashboard, tenant grievance portal and role-based access (super admin, owner, warden).
- Layered FastAPI backend with JWT, rate limiting, Alembic migrations and indexed PostgreSQL queries for rent and electricity invoicing, partial payments and defaulter tracking; covered by 73 pytest tests plus Playwright E2E tests.

Blockchain E-Voting System | Solidity, Smart Contracts, Web3.js

- Led a 6-member team to the Smart India Hackathon 2022 Grand Finale from 2,000+ teams.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Go, C

Backend: FastAPI, Django, Flask, Node.js, SQLAlchemy, REST API design, microservices, JWT, RBAC, rate limiting

Databases: PostgreSQL, Redis, DynamoDB, schema design, Alembic migrations

Cloud and DevOps: AWS (S3, CloudWatch), Docker, Docker Compose, GitLab CI/CD

Testing: pytest, Jest, Playwright (E2E), API testing (Postman)

Frontend: React, Next.js, Tailwind CSS, HTML, CSS

Tools and practices: Git, GitHub, GitLab, Firebase Cloud Messaging, secret scanning, Agile, Scrum

AI-Assisted Development: Claude, Cursor, Amazon Q Developer, Kiro

EDUCATION

Bachelor of Technology in Computer Technology | Bannari Amman Institute of Technology

November 2021 - April 2025 | CGPA: 7.96/10.0